

RITIK SARRAF

Individual Project

Global AI Job Market and Salary Trends

- ▶ Dataset: [Kaggle](#) with 15000 rows and 20 different categorical variables (columns)
- ▶ Source: Global AI Job Markets and Salary Trends (
- ▶ Observations: A Comprehensive Analysis of Global Employment Trends in AI Synthetically
- ▶ Time Period: 2025

Industry Types

- ▶ Total of 15 leading Industry types.
- ▶ Check to observe what industry types has greater effect on employee's income.
- ▶ What factors contributes to the impact on Income?

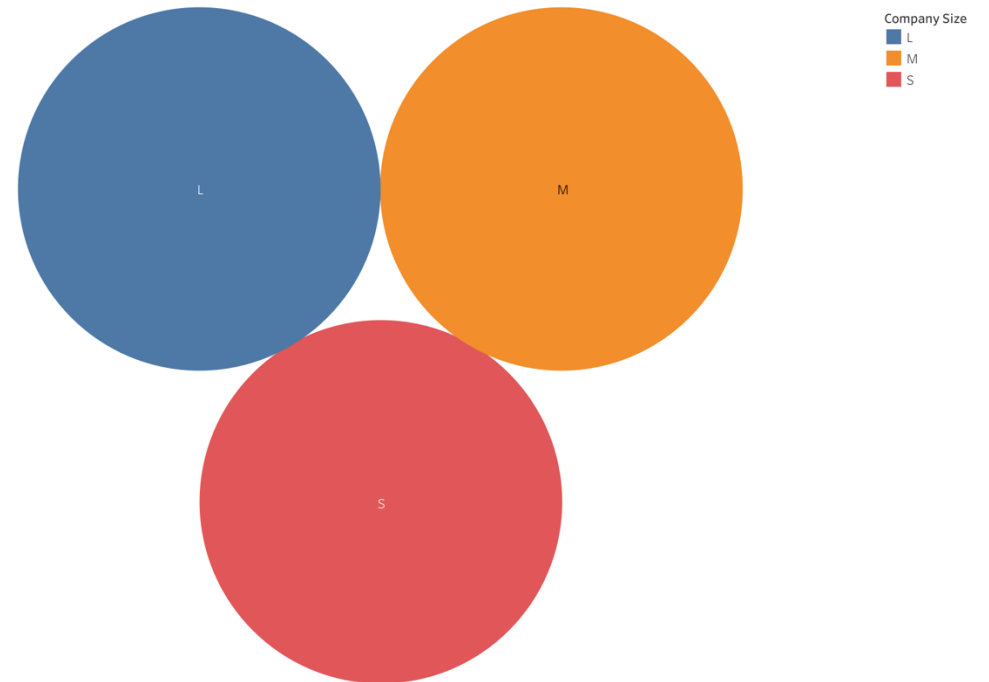
Industry



Industry Scale

- ▶ Large Scale (L)
- ▶ Medium Scale (M)
- ▶ Small Scale (S)

Industry Scale



Company Name

- ▶ Total of 15 Companies
- ▶ Check to observe what industry types has greater effect on employee's income
- ▶ What factors contributes the most and the least to the impact on Income?

Company Name



Company Location

- ▶ Companies are based in altogether 20 different countries.
- ▶ Makes it diverse as covers four different continents.
 - ▶ Asia
 - ▶ Europe
 - ▶ North America
 - ▶ Australia

Company Location



Job Title

- ▶ Including all Industries and Companies, dataset contains for job markets in 20 different sectors.
- ▶ Opening of job markets, as dataset contains for employee's income status from diverse backgrounds:
 - ▶ Scientists
 - ▶ Consultant
 - ▶ Engineer
 - ▶ Analyst

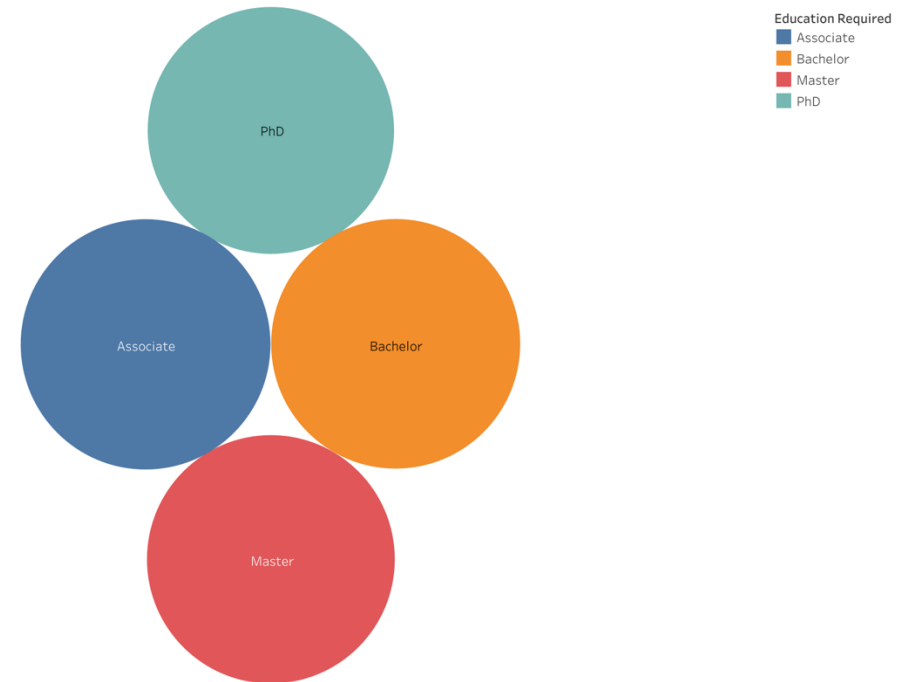
Job Title



Education Degree

- ▶ Associate Degree
- ▶ Bachelor Degree
- ▶ Master's Degree
- ▶ PhD

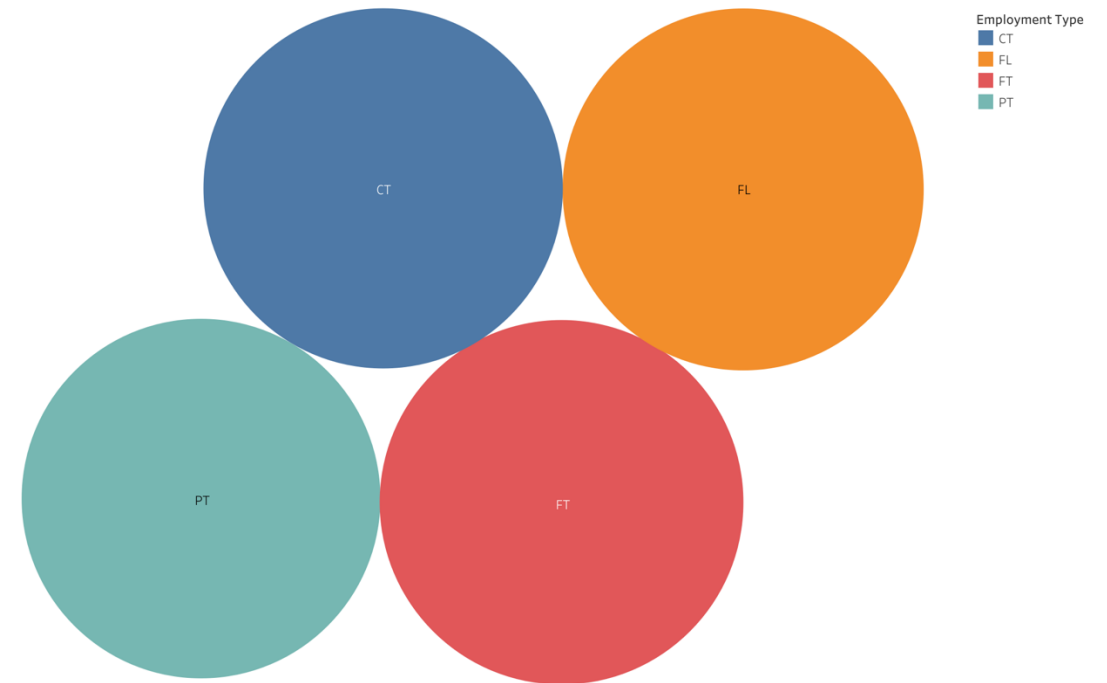
Education Degree Level



Employment Type

- ▶ Freelance (FL)
- ▶ Contract (CT)
- ▶ Part-Time (PT)
- ▶ Full-Time (FT)

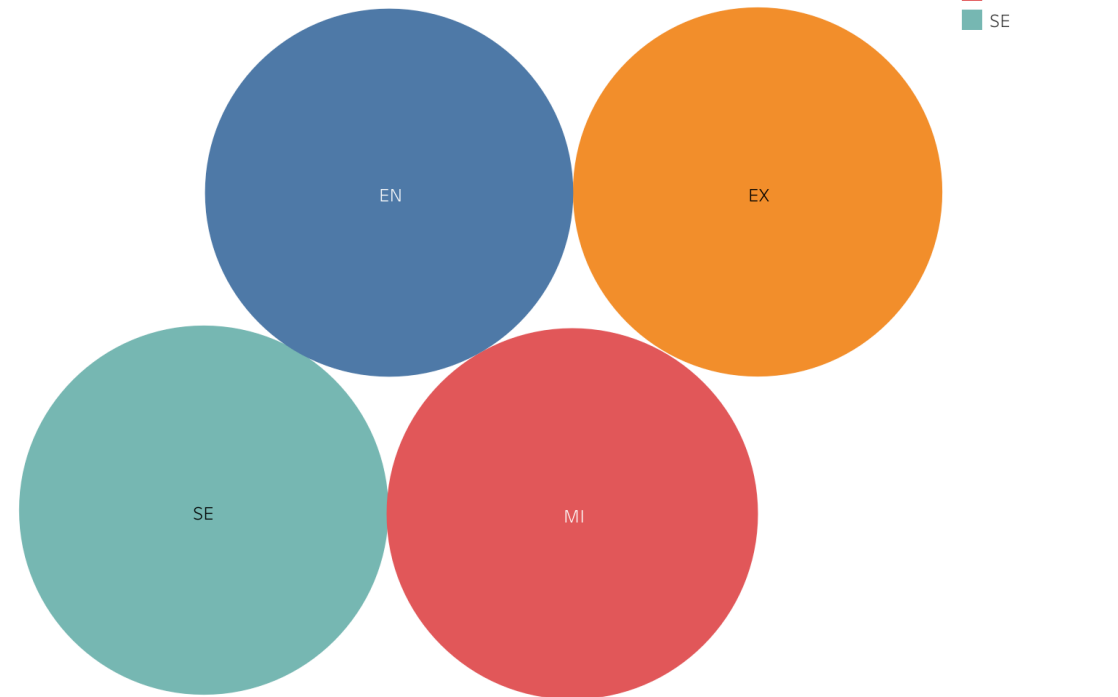
Employment Type



Experience Level

- ▶ Entry Level (EN)
- ▶ Middle Level (MI)
- ▶ Senior Level (SE)
- ▶ Executive Level (EX)

Experience Level



Employee Residence

- ▶ Employees are based in altogether 20 different countries.
- ▶ Makes it diverse as covers four different continents.
 - ▶ Asia
 - ▶ Europe
 - ▶ North America
 - ▶ Australia

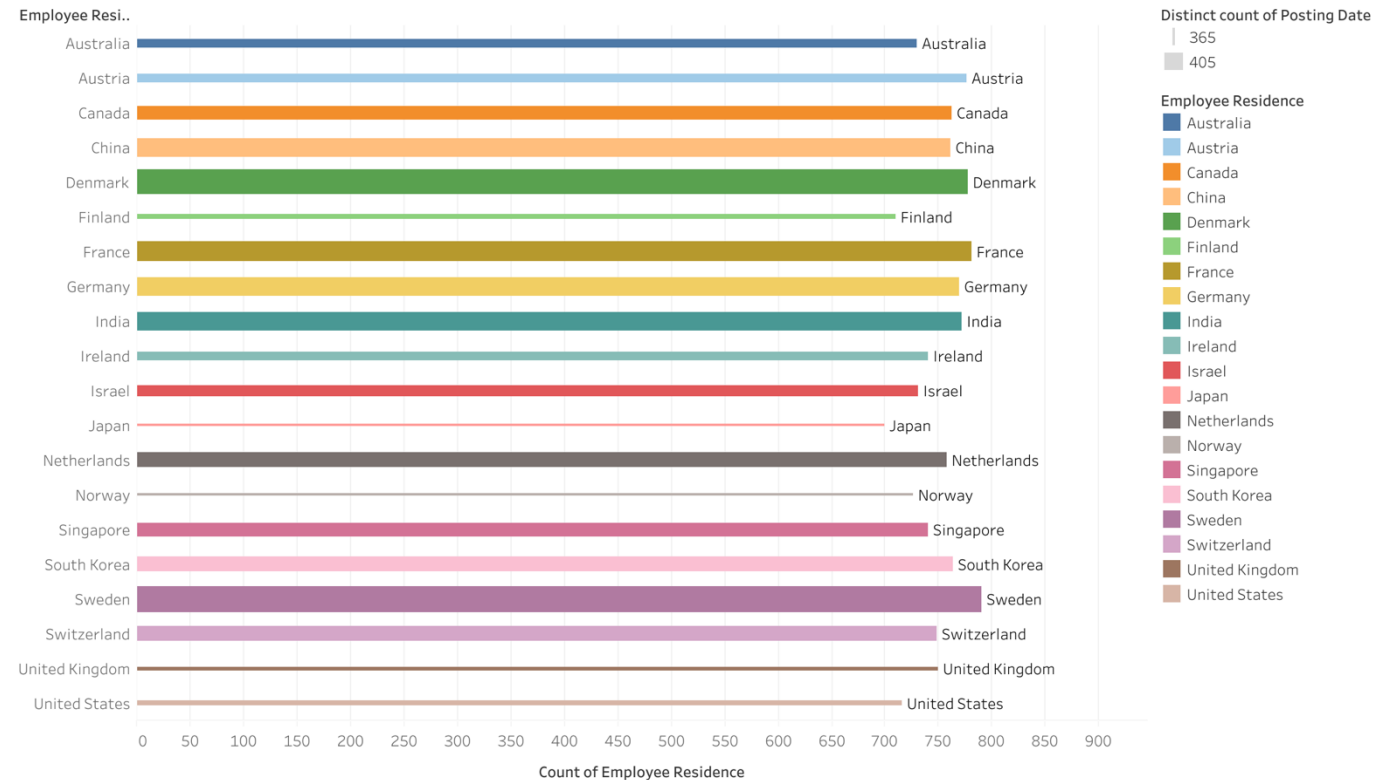
Employee Residence



Employee Residence Count

► Employee represents most from Sweden with the population of 790 and the least from Japan with the population of 700.

Employee Residence Count

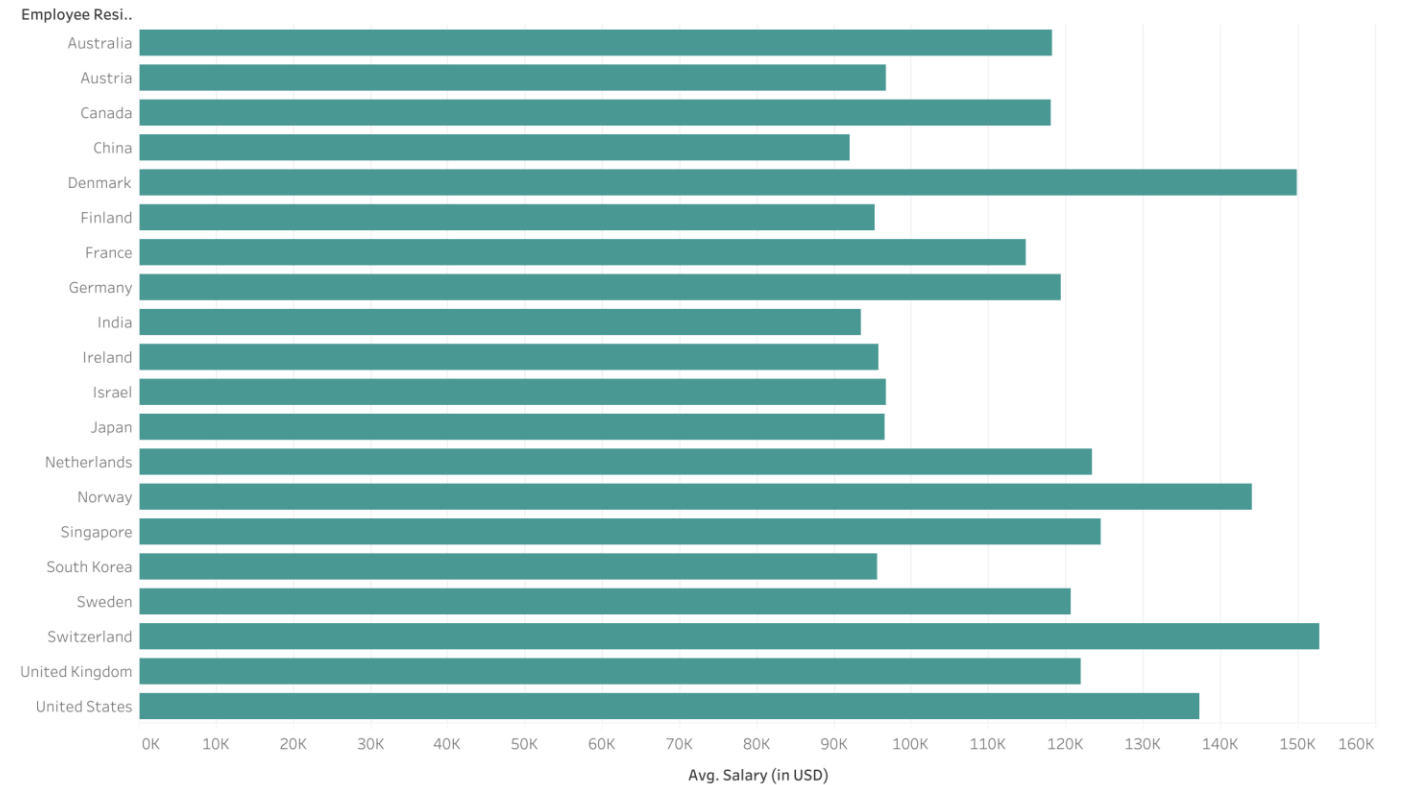


Average Salary by Residence

► Employees from Switzerland and Denmark are among the highest average paid within the industry.

► Employees from China are among the least average paid.

Avg. Salary Distribution by Employee Residence

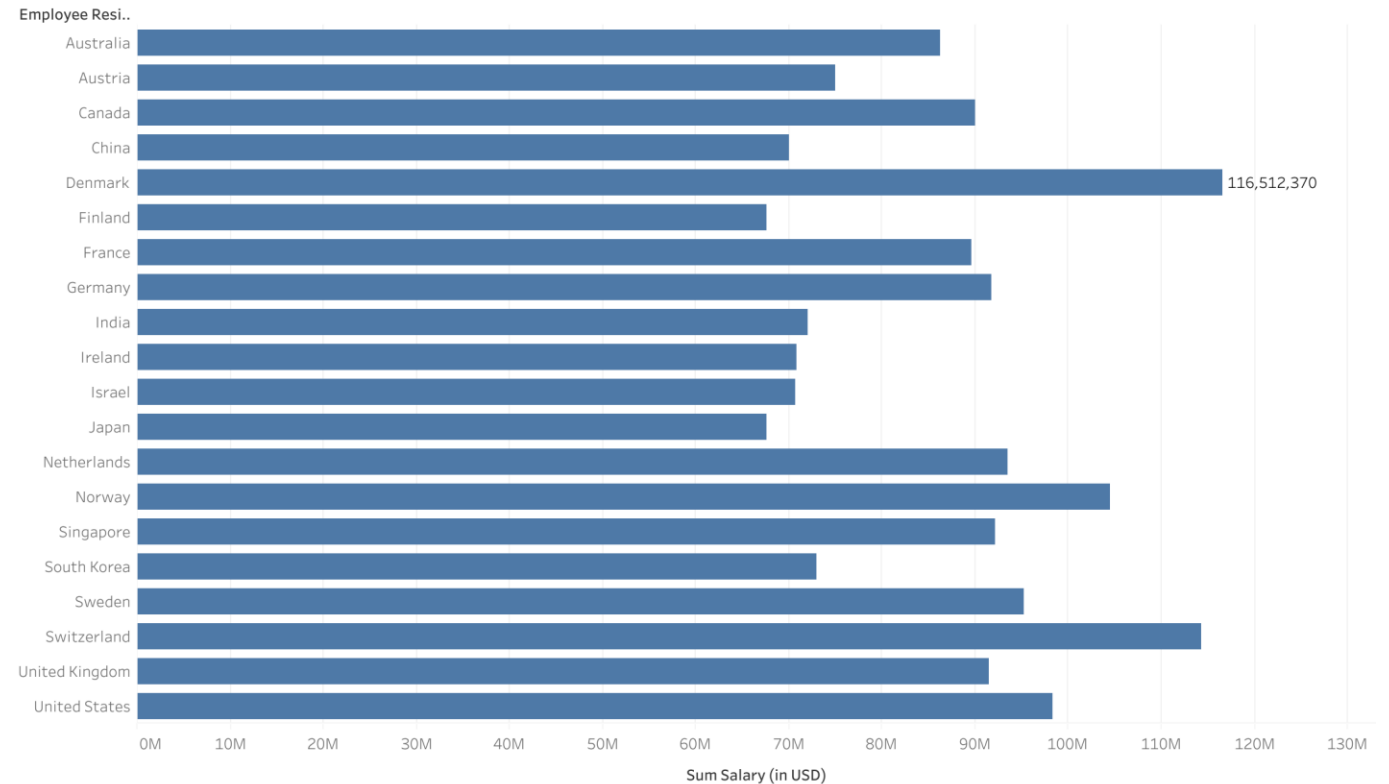


Total Salary by Residence

► Employees from Switzerland and Denmark are among the highest cumulative salaried within the industry.

► France and Japan are among the least cumulative salaried employees.

Sum Salary Distribution by Employee Residence

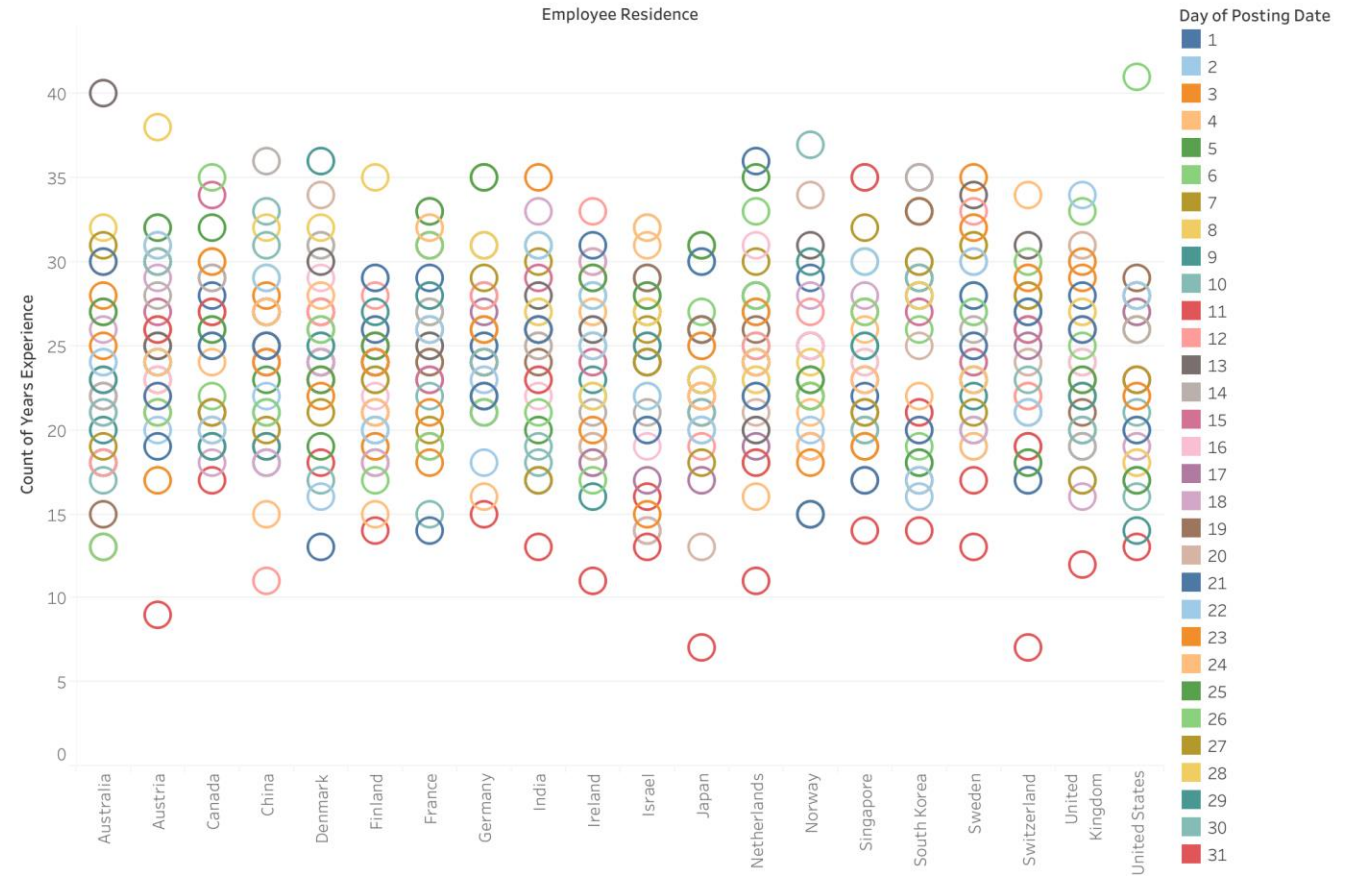


Years of Experience by Residence

► Denmark, France, Israel, Canada, UK including some other countries represents lower variance in Experience.

► Some countries has outliers such as USA, Australia, Japan, Switzerland, etc.

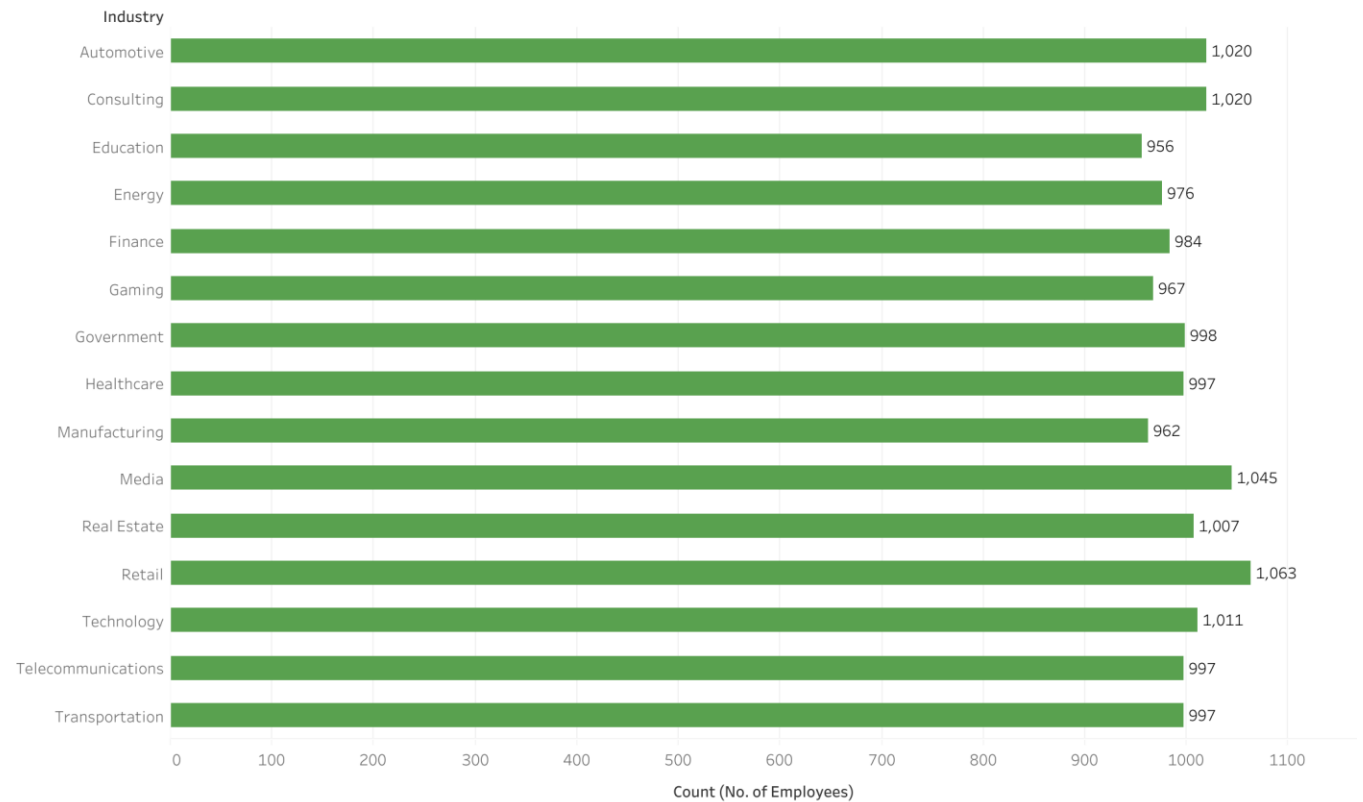
Years of Experience (Scatter)



Industry Type Count

- ▶ Employee represents most from Retail Industry with the population of 1063
- ▶ Least from Education with the population of 956.

Industry Type Count



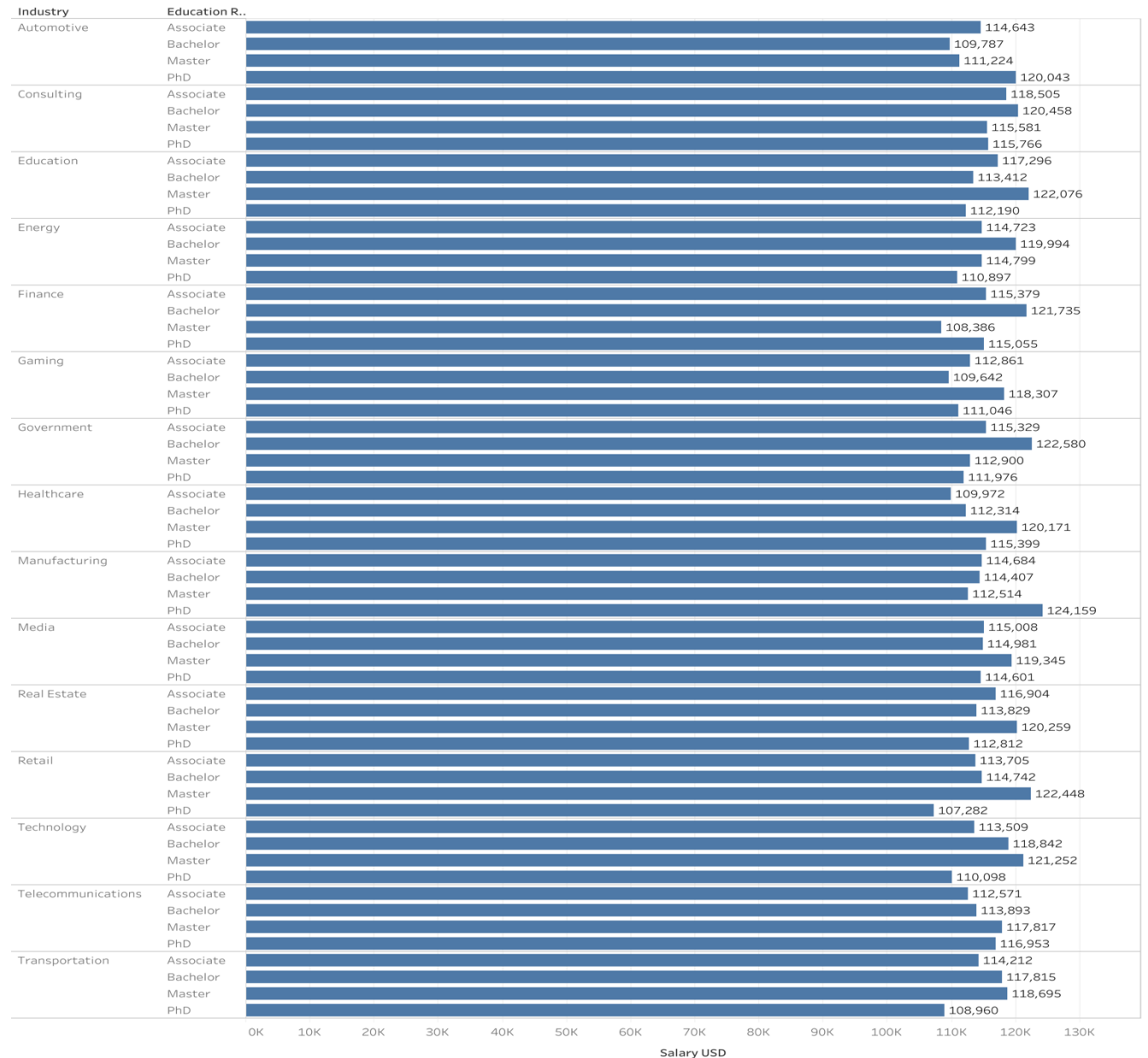
Average Salary by Industry

► PhD Employees from Automotive and Manufacturing are among with highest average salary.

► Bachelor's Employees from Consulting, Energy, Finance and Government are among the highest average salary.

► Master's Employees from rest of nine other sectors are among the highest average salary.

Avg. Salary Distribution by Industry



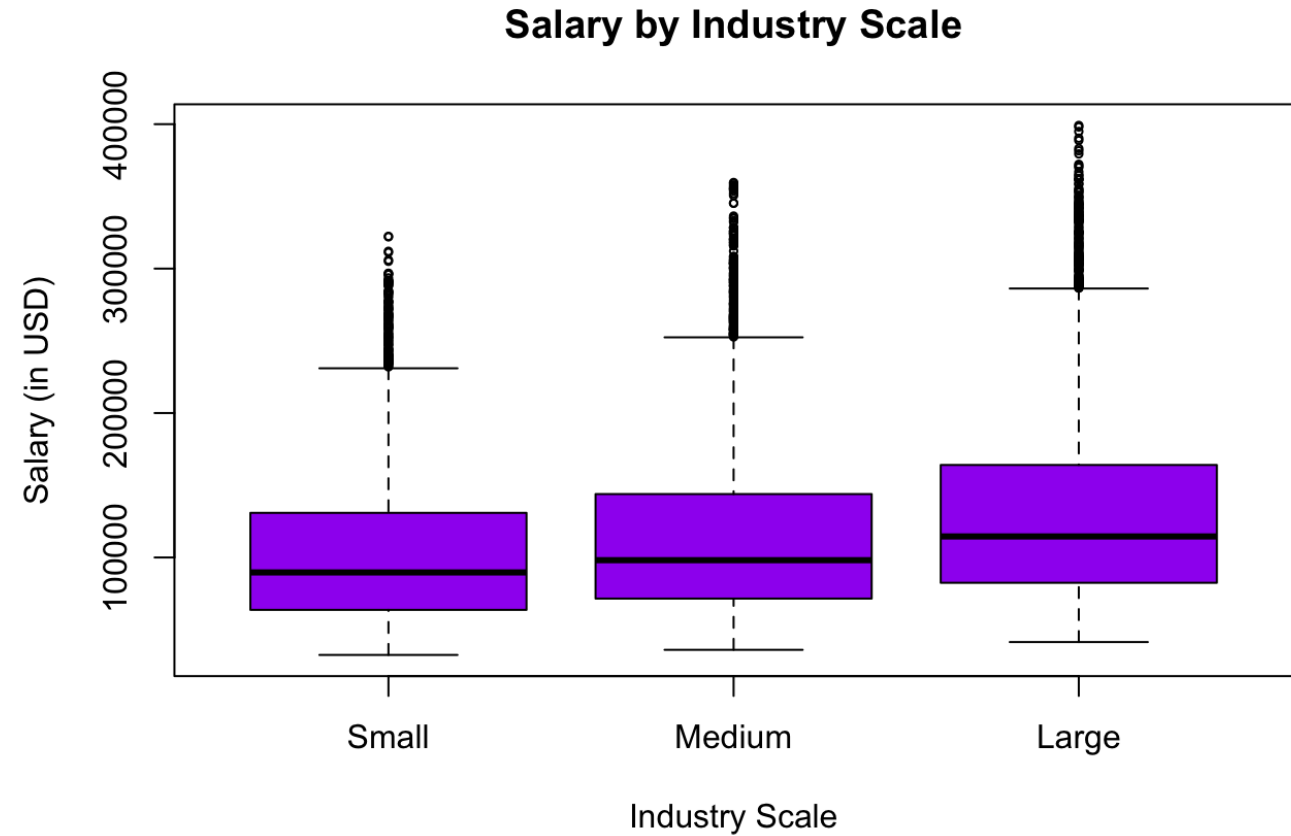
Median Salary by Industry

Highest Median Salary

► Large Scale

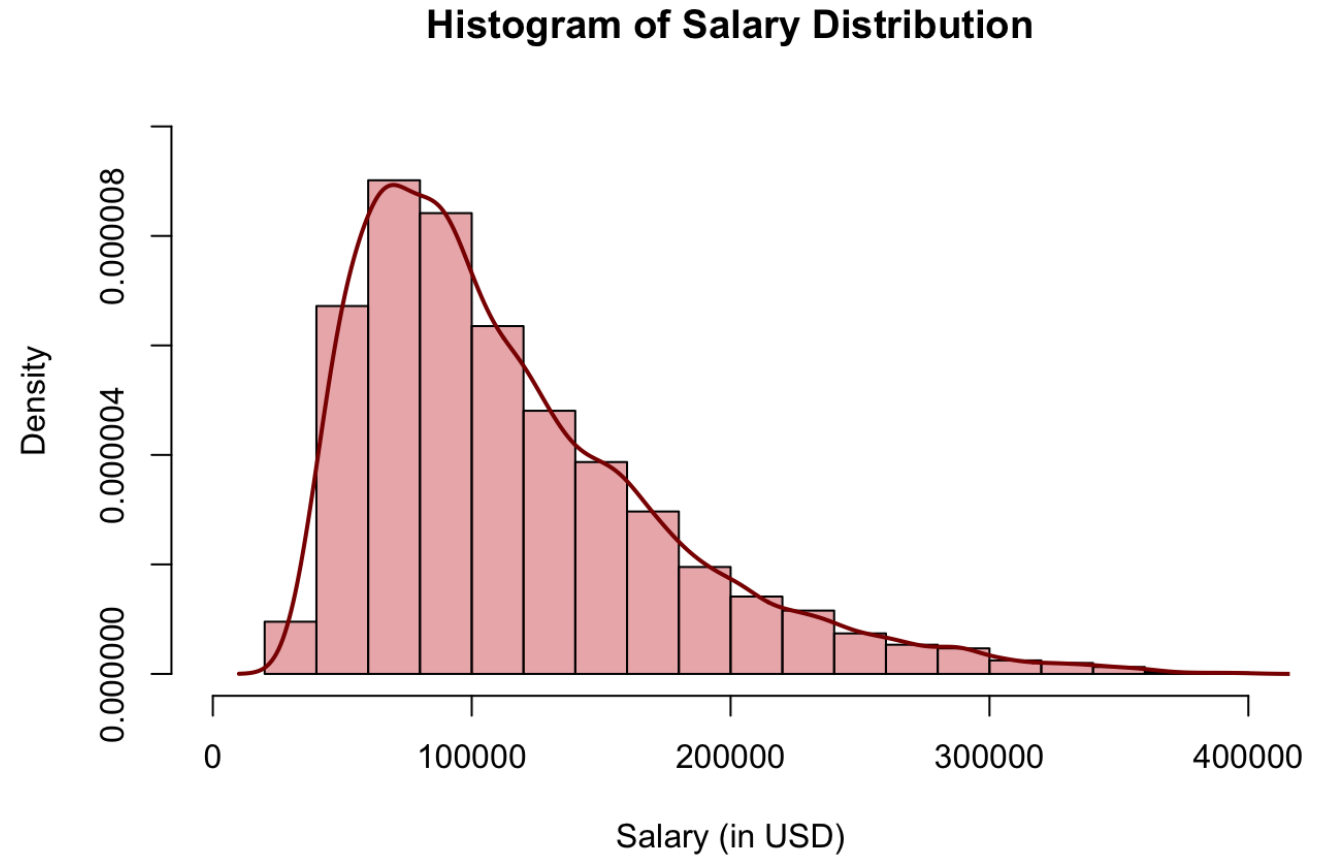
Lowest Median Salary

► Small Scale



Salary Distribution

- ▶ Positive Skewed Histogram
- ▶ More employees have salary in range between \$50,000 and \$100,000
- ▶ Number of employees decreases as the salary range gets higher.

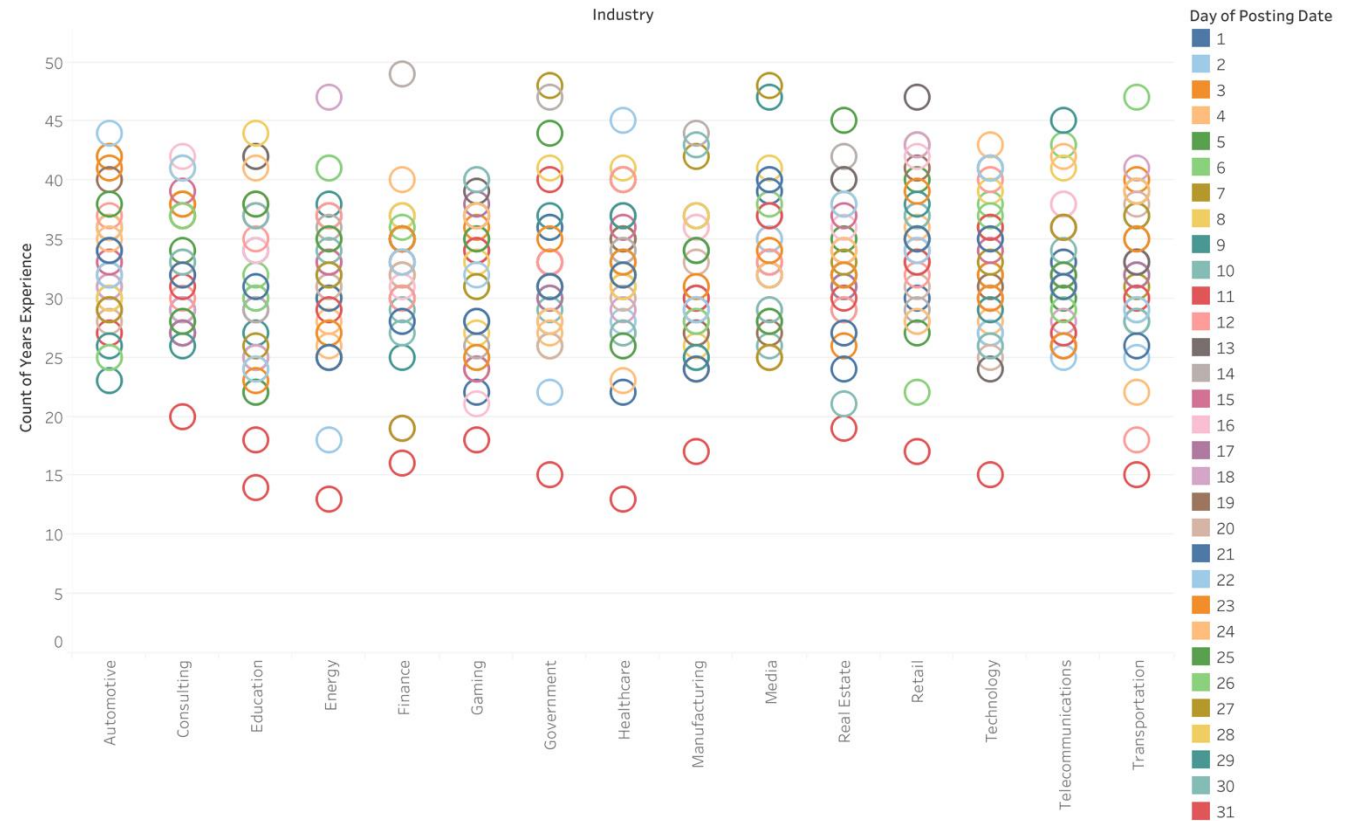


Years of Experience by Industry

► Employees from Automotive and Telecommunication Industry has least fluctuation on experience.

► Employees from Transportation, Energy and Finance industry are among the highest fluctuation on experience.

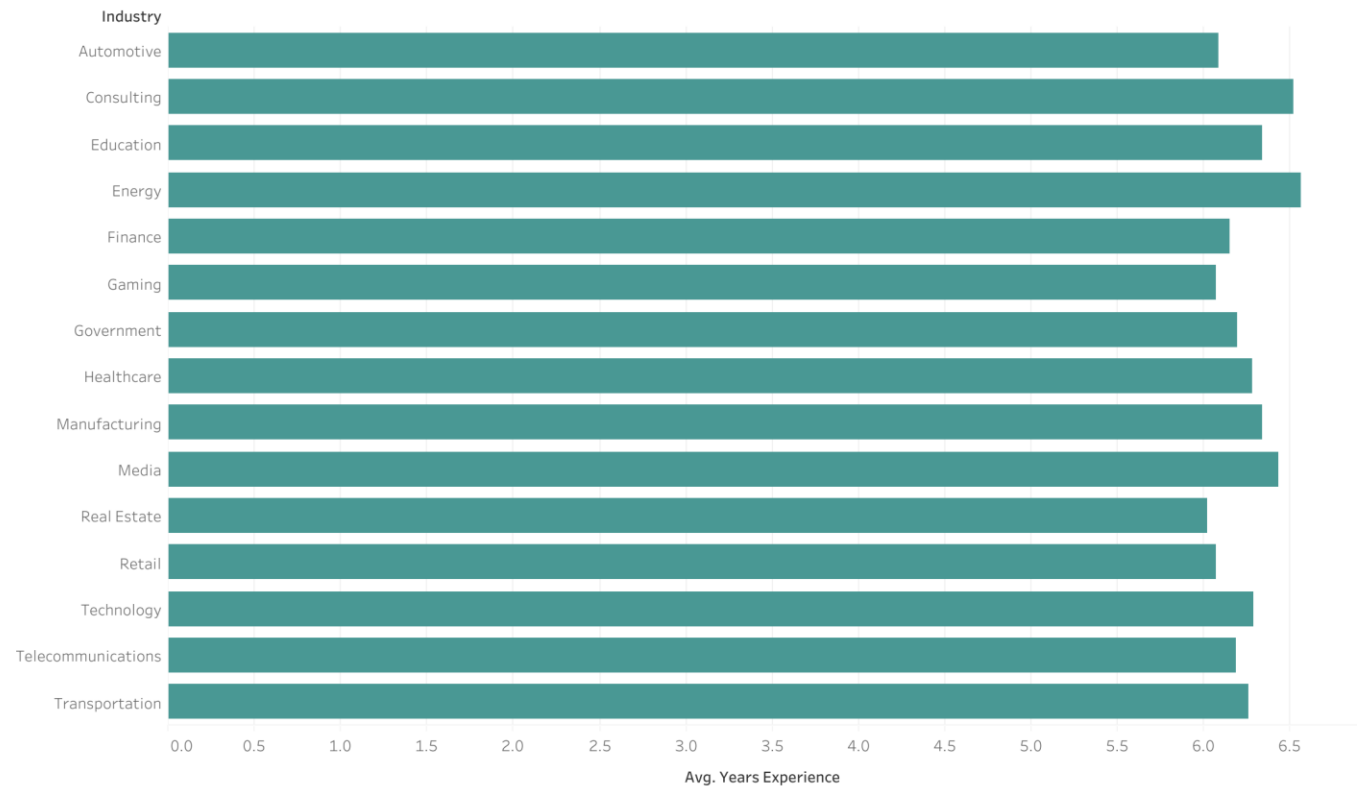
Years of Experience (Scatter)



Average Experience by Industry

- ▶ Employees from Education Industry have longer average experience.
- ▶ Employees from Real Estate industry are among the low average experienced.

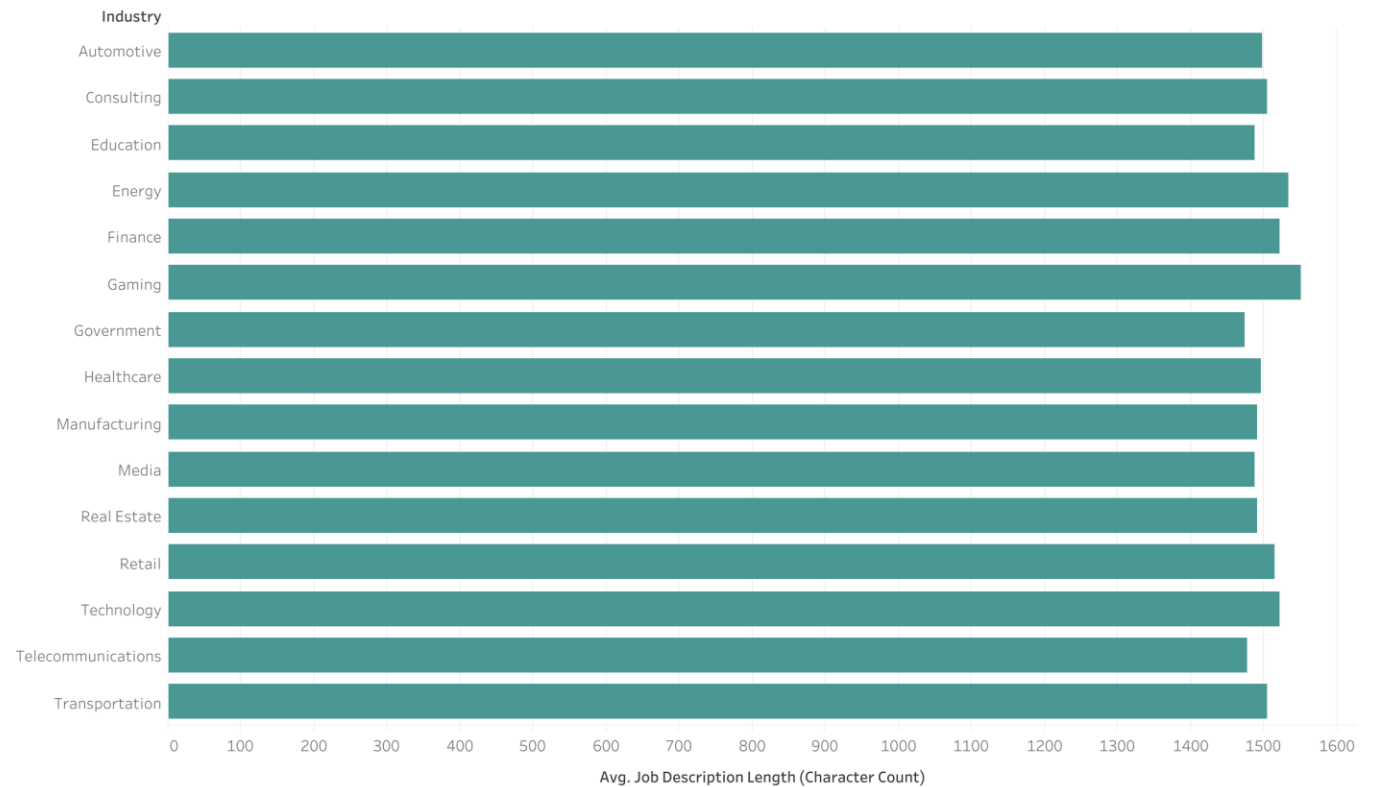
Average Years of Experience by Industry Type



Average Job Description Length by Industry

- ▶ Employees from Government Industry has on average fewer job duties.
- ▶ Employees from Gaming industry has on average higher job duties to be fulfilled.

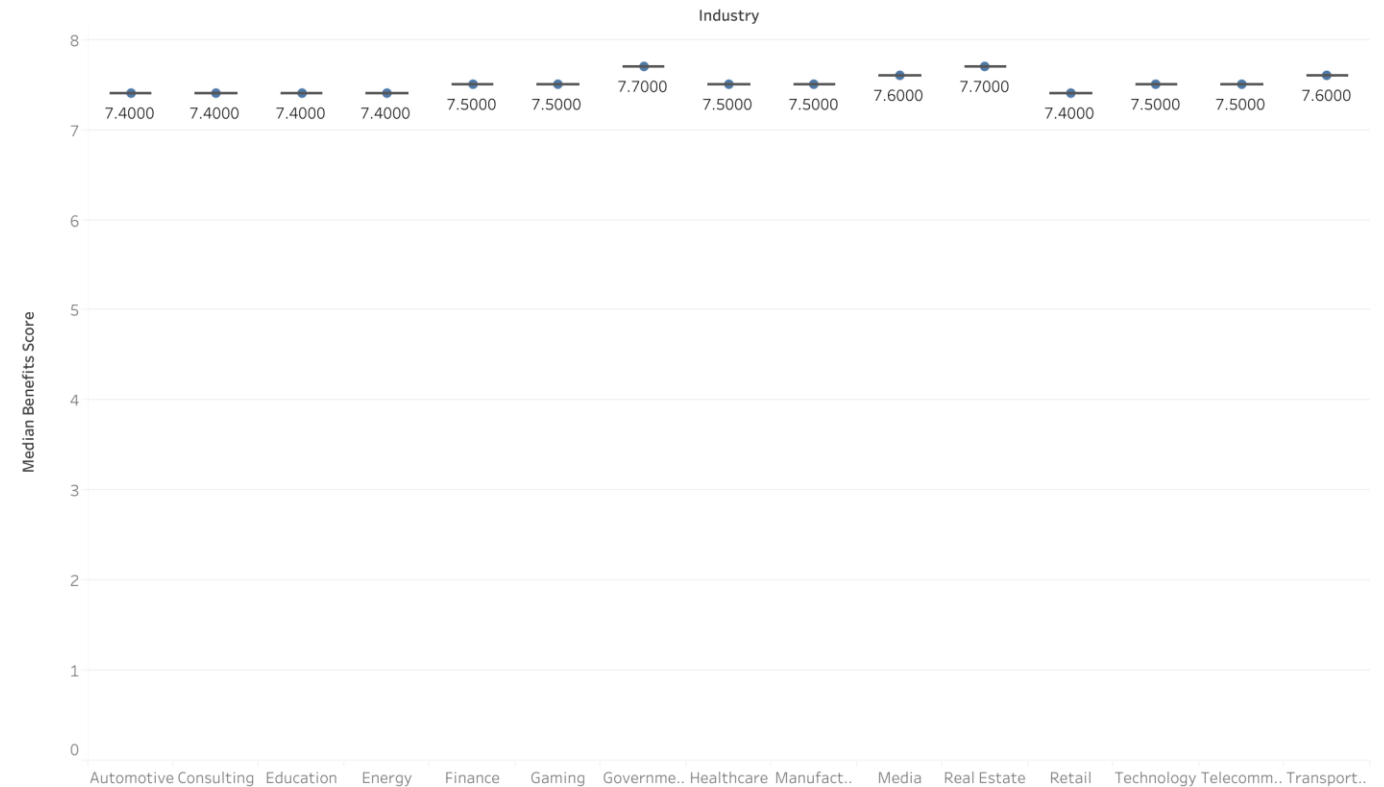
Avg. Job Description Length by Industry Type



Median Benefits Score by Industry

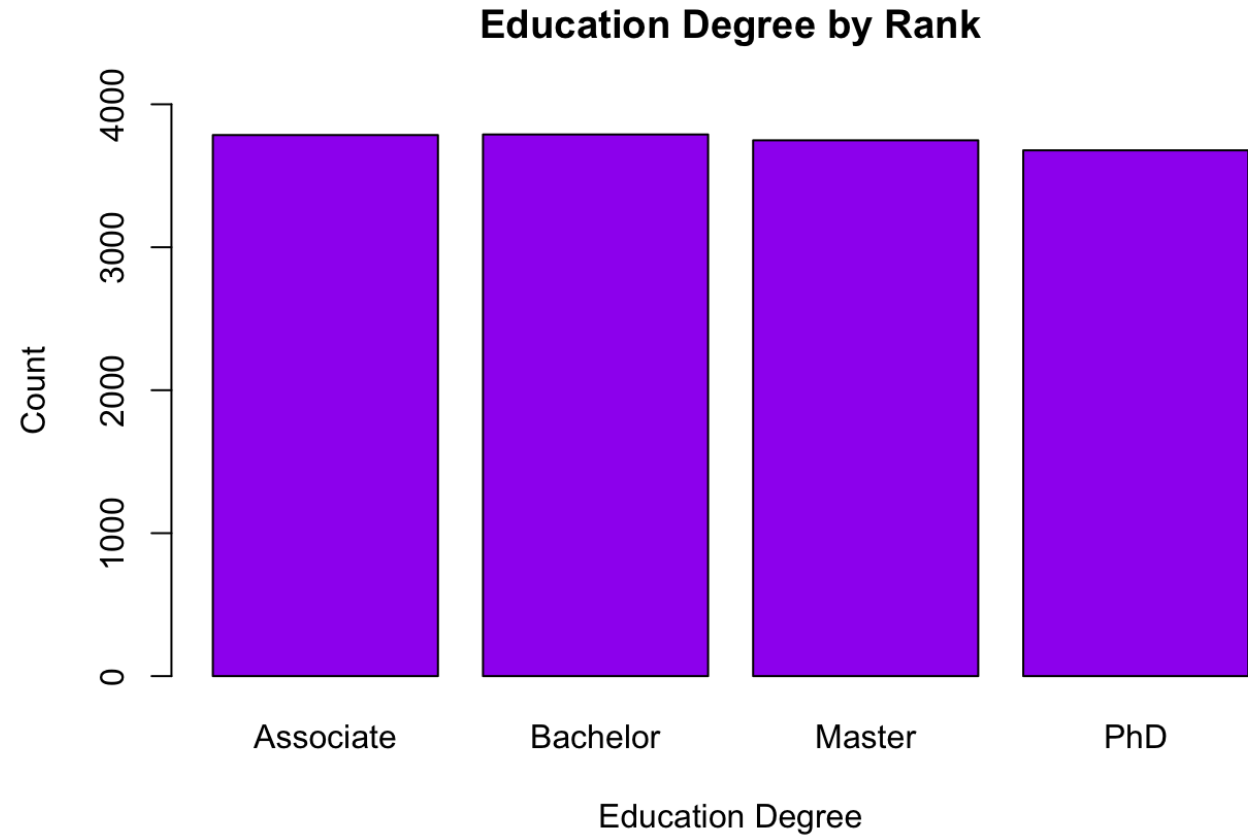
- ▶ More than 50% employees from Real Estate and Government sectors enjoy benefits of insurance, social security, Savings plan (401K), etc.
- ▶ This make employees and/or family members happy resulting in better quality of life.

Median Benefits Score by Industry



Education Degree Count

- ▶ Bachelor's
- ▶ Associate
- ▶ Master's
- ▶ PhD



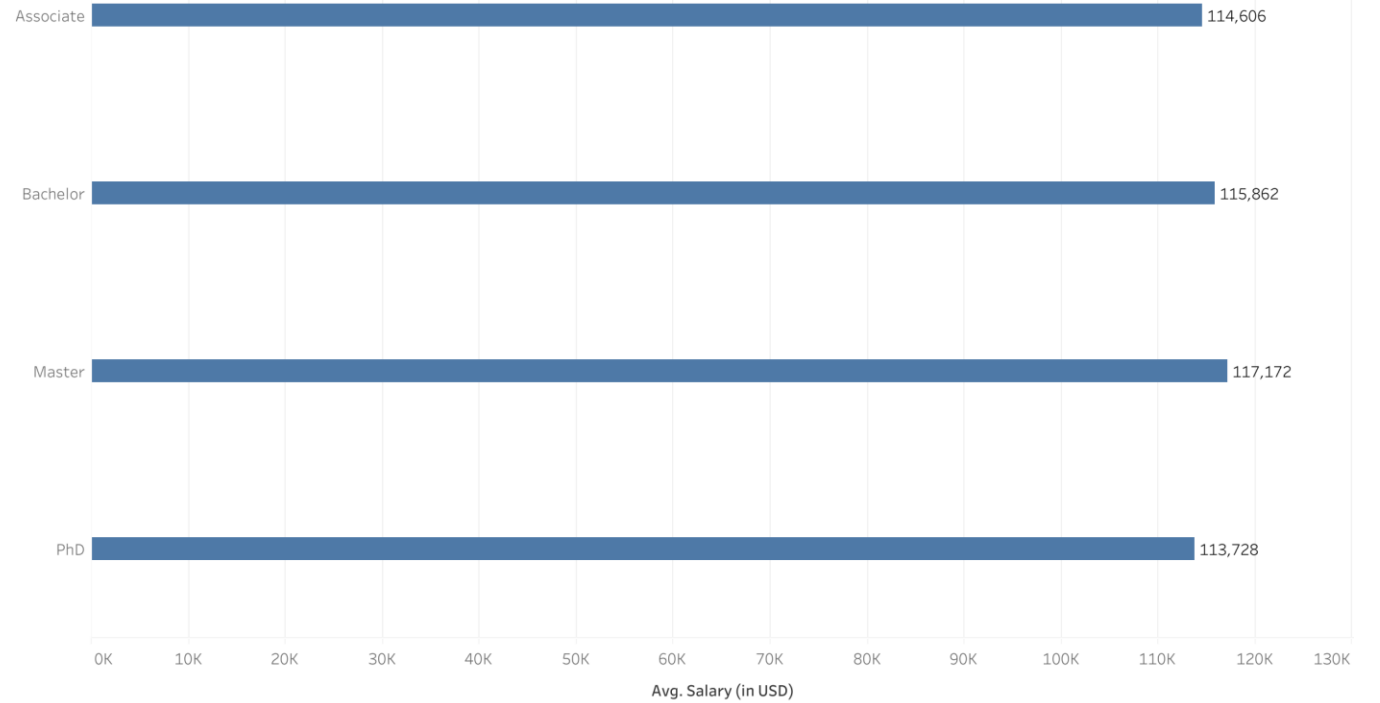
Average Salary by Education Degree

- ▶ Master's Degree
- ▶ Bachelor's Degree
- ▶ Associate Degree
- ▶ PhD



Average Salary by Education Degree

Education R..

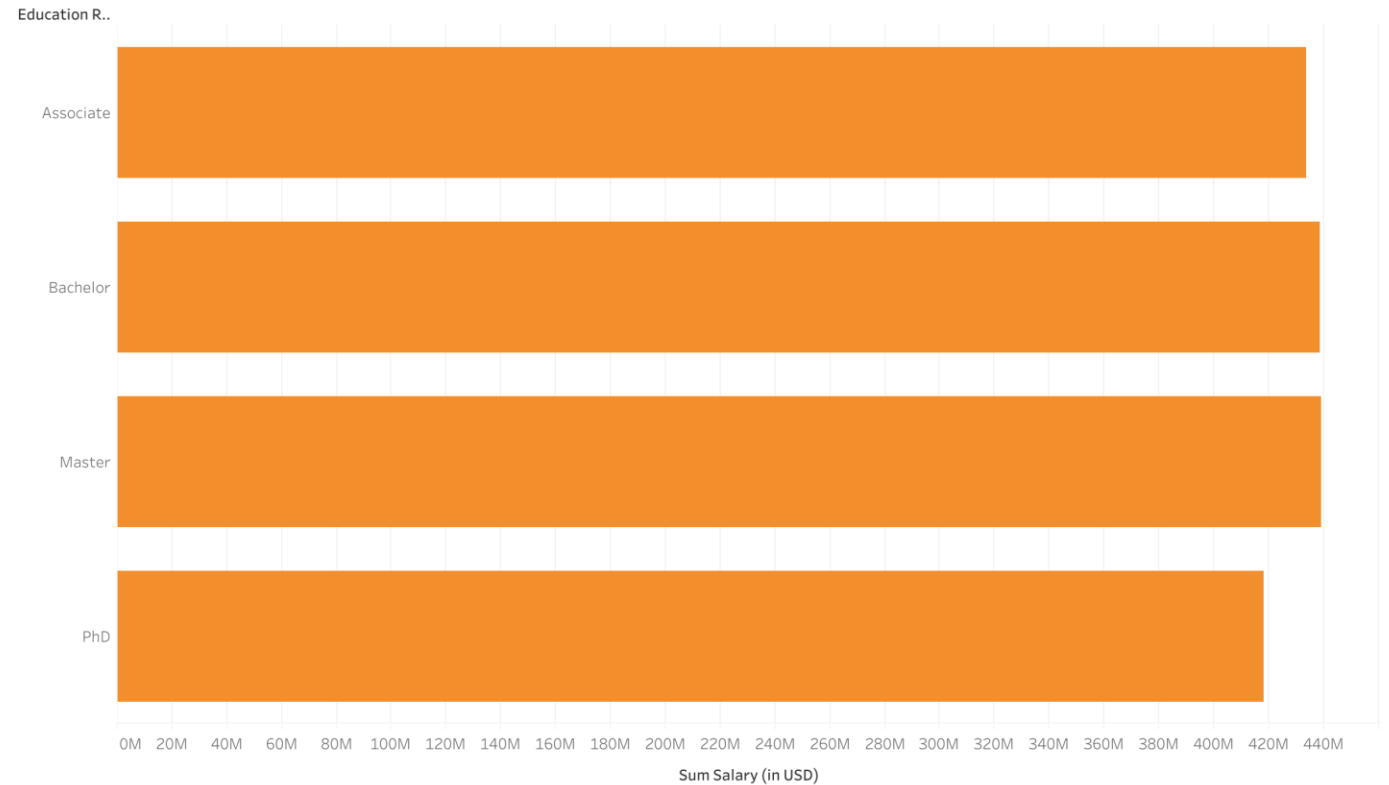


Sum Salary by Education Degree

- ▶ Master's Degree
- ▶ Bachelor's Degree
- ▶ Associate Degree
- ▶ PhD



Sum Salary Distribution by Education Degree

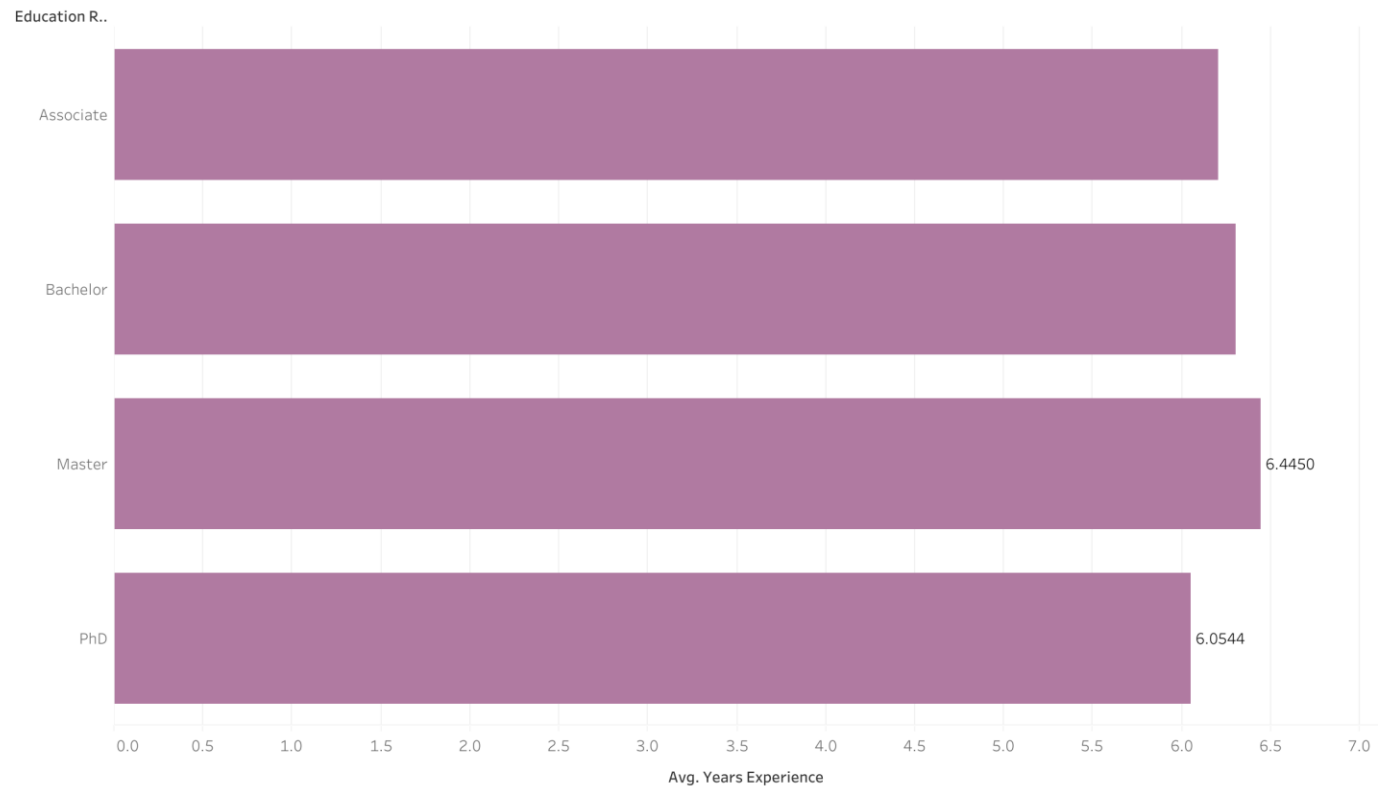


Average Experience by Education Degree

► Employees with Master's have on average longer experience.

► Employees with PhD have on average shorter experience.

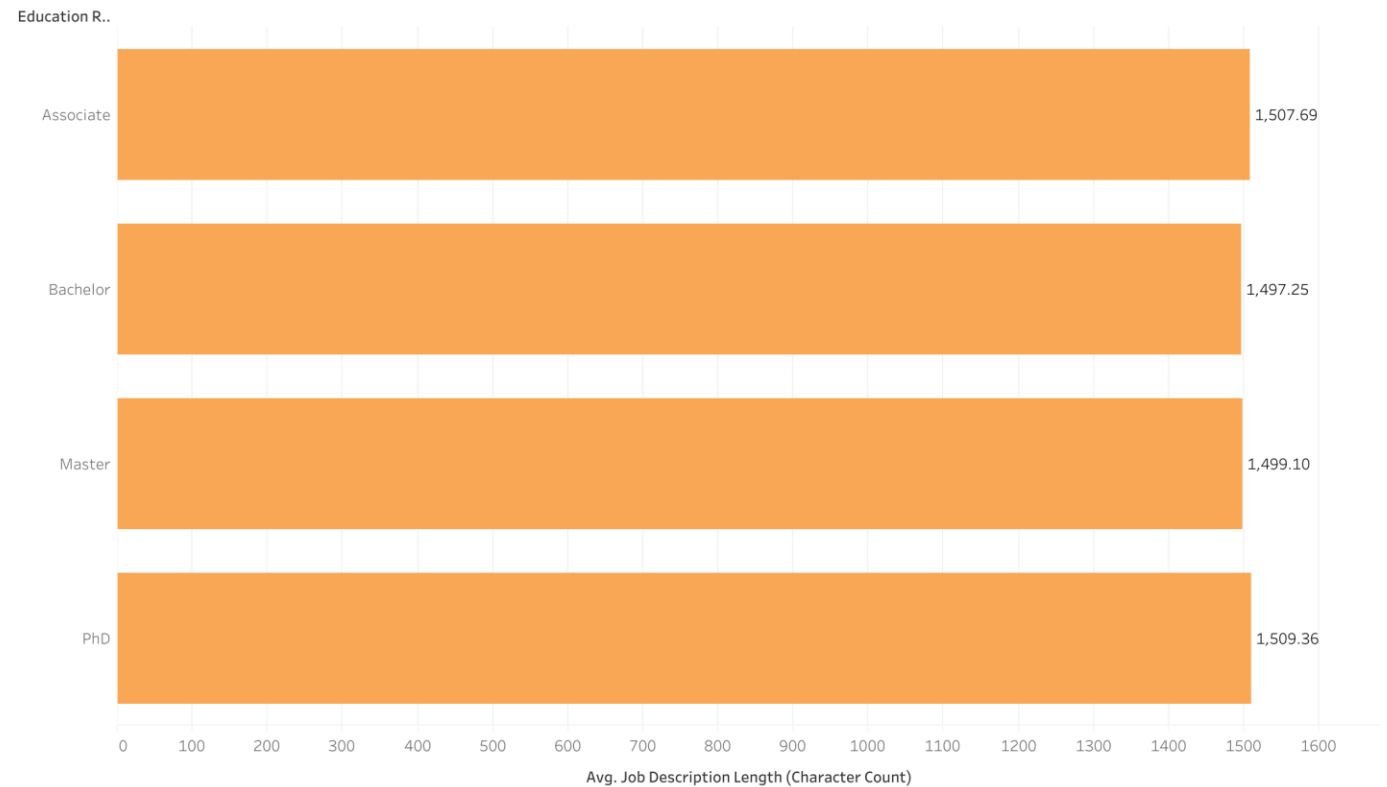
Avg. Years of Experience by Education Degree



Avg. Job Description Length by Education Degree

- ▶ PhD holders has on average higher job duties.
- ▶ Bachelor's Degree holders has on average lesser job duties.

Avg. Job Description Length by Education Degree

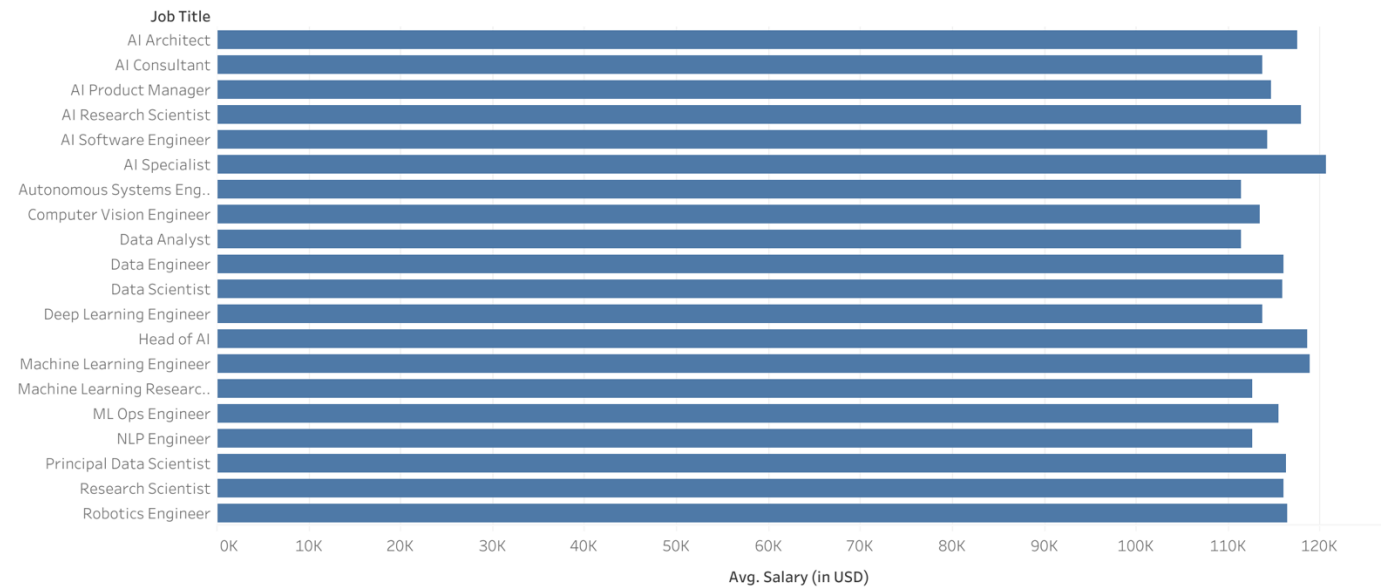


Average Salary by Job Title (Top 5)

- ▶ AI Specialist
- ▶ Machine Learning Engineer
- ▶ Head of AI
- ▶ AI Research Scientist
- ▶ AI Product manager



Avg. Salary Distribution by Job Title



Median Salary by Employment Level

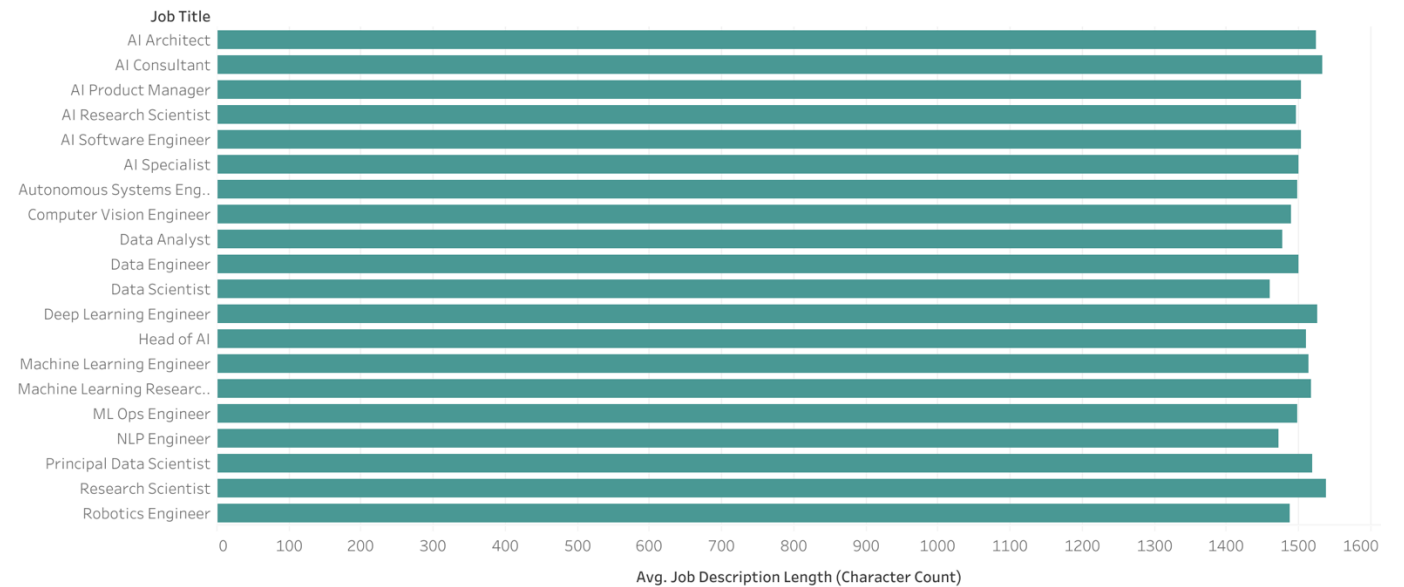
- ▶ Executive Level employee has highest median salary with greater variance.
- ▶ Entry Level employee has lowest median salary with least variance.



Average Job Description Length by Job Title

- ▶ Research Scientists have on average more job duties.
- ▶ Data Scientists have on average fewer job duties to be fulfilled.

Avg. Job Description Length by Job Title



Avg. Job Description Length vs. Salary by Job Title

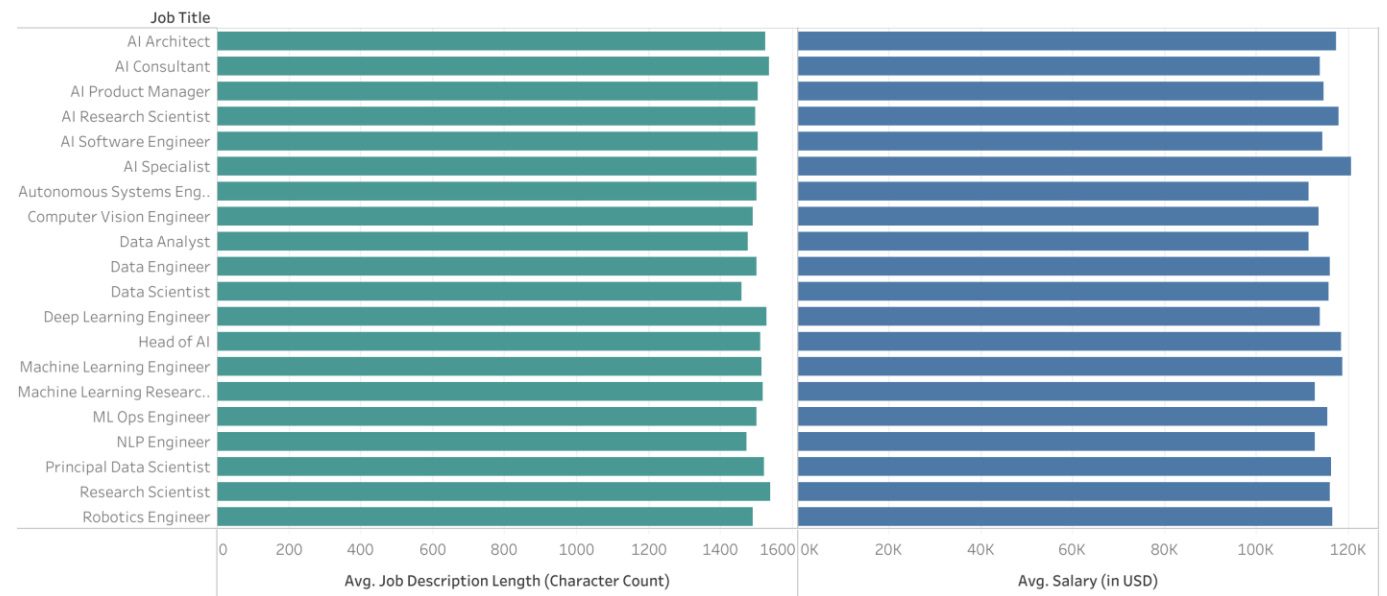
► Job Description Length

Research Scientists > Data Scientist

► Salary

Data Scientist > Research Scientist

Avg. Job Description Length vs. Avg. Salary Distribution by Job Title

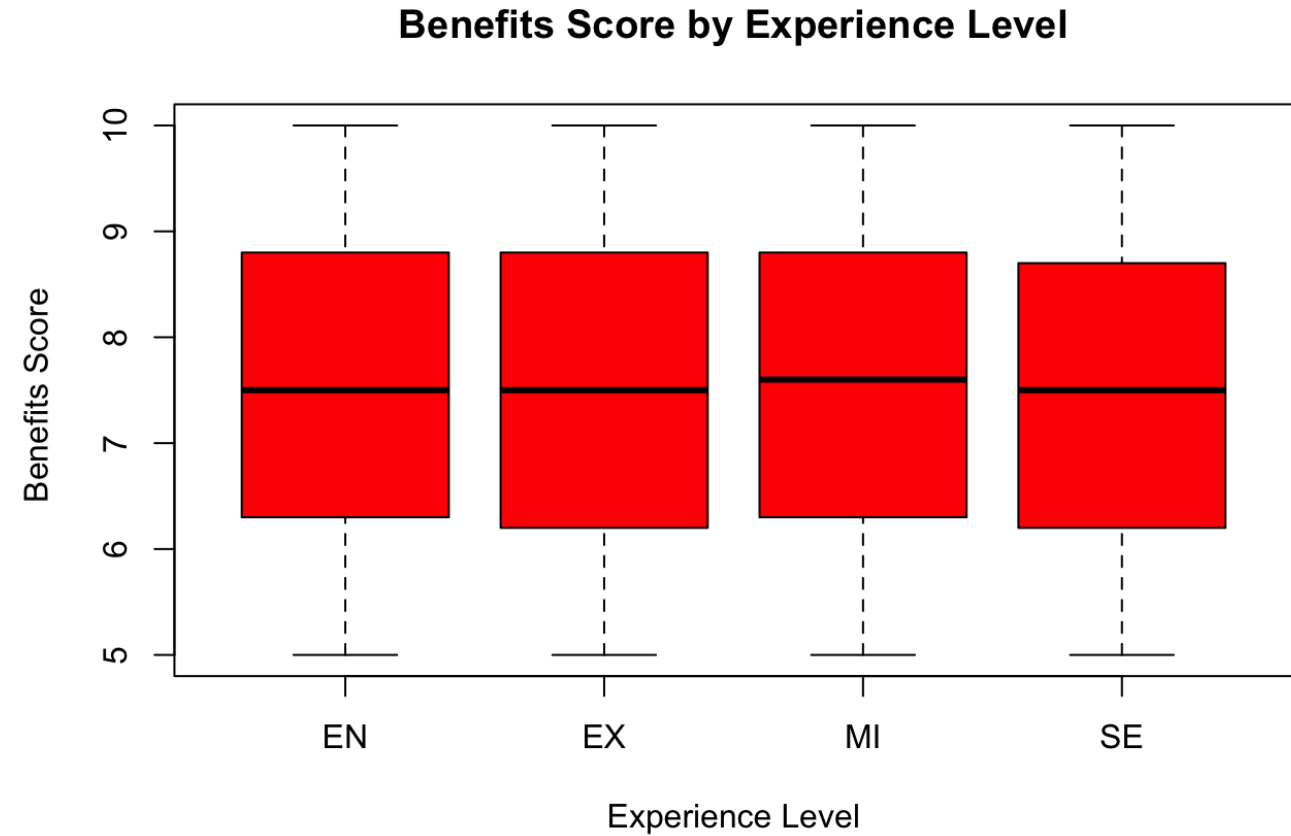


Avg. Job Description Length vs. Salary by Experience

Highest Median Benefits Score

► Middle Level

~ No Significant Effect



Conclusion

- ▶ Job Length Description and more duties does not clarify about higher pay rate.
- ▶ Higher Degree Education does not clarify for higher salary compensation on all industry types.
- ▶ Government (federal and state) and Real Estate sectors have higher benefits score.
- ▶ Salary Compensation in two different jobs sector are infeasible to compare based on experience level.
- ▶ Higher Degree Education corresponds to more work and duties to handle.
- ▶ Higher Job postings are positively correlated with salary distribution.

Thank you! 😊